

# Well2Surface Digital Twin: Execution Specification

AI Collaboration Routing Matrix: Claude Sonnet 5 vs. Gemini (Flash / Pro)

*Asset Focus: Heavy Oil Wells of Baghewala Field (CSS & SRP Operations)*

## Model Allocation Strategy

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To optimize resource allocation and model capabilities, development tasks are split between **Gemini Flash 3.6 / 3.1 Pro** and **Claude Sonnet 5** based on computational complexity, mathematical rigor, and architectural risk:

- **Gemini Flash 3.6 (Trivial / Low):** Handles repetitive scaffolding, static configurations, basic algebra, serializations, and templated exports.
- **Gemini Pro 3.1 (Low–Med / Med):** Solves single-domain formulas, multi-variable logic, UI dashboards, feature transformers, and edge-case unit testing.
- **Claude Sonnet 5 (High / Very High):** Reserved for non-convex optimization, coupled PDEs/ODEs, stability proofs, multiphase hydraulic solvers, and end-to-end integration architecture.

Task Difficulty & AI Assignment Matrix

| Task Description                                                      | Difficulty | Assigned Model  | Effort Level      | Engineering Rationale & Notes                                       |
|-----------------------------------------------------------------------|------------|-----------------|-------------------|---------------------------------------------------------------------|
| <i>Phase 0 — Scaffold &amp; Config</i>                                |            |                 |                   |                                                                     |
| Repo Scaffold (dirs, <code>__init__.py</code> )                       | Trivial    | Gemini Flash    | –                 | Pure boilerplate, zero judgment.                                    |
| <code>field_params.yaml</code> / <code>operational_limits.yaml</code> | Trivial    | Gemini Flash    | –                 | Static parameter entry, no logic.                                   |
| <code>requirements.txt</code>                                         | Trivial    | Gemini Flash    | –                 | Package dependency list.                                            |
| <i>Phase 1 — CSS Thermal EOR Engine</i>                               |            |                 |                   |                                                                     |
| $Q_{inj}$ Steam Enthalpy Equation                                     | Low–Med    | Gemini Pro      | –                 | Validates $x_{qual}$ bound-check logic.                             |
| $Q_{absorbed}$ Composite Heat Capacity                                | Low–Med    | Gemini Pro      | –                 | Multi-term algebraic formula.                                       |
| Thermal Decay ODE + Dynamic $\alpha$                                  | High       | Claude Sonnet 5 | High Effort       | Coupled production terms; unit dynamics ( $s^{-1}$ vs $day^{-1}$ ). |
| Andrade Viscosity $\mu(t)$ Calibration                                | Med        | Gemini Pro      | –                 | Calibrates $A/B$ against target cP data.                            |
| Safety Guardrails ( $T < T_{res}$ , $\mu \leq 0$ )                    | Trivial    | Gemini Flash    | –                 | Direct exception raise logic.                                       |
| <i>Phase 2 — SRP Mechanical Lift Engine</i>                           |            |                 |                   |                                                                     |
| Rod Stretch (Hooke’s Law)                                             | Trivial    | Gemini Flash    | –                 | One-line formula.                                                   |
| Viscous Drag Force $F_{drag}$                                         | Low        | Gemini Pro      | –                 | Log-ratio term, unit consistency ( $Pa \cdot s$ vs cP).             |
| Buoyant Weight $W_{buoy}$                                             | Trivial    | Gemini Flash    | –                 | Direct hydrostatics formula.                                        |
| PPRL + Yield Interrupt Flag (> 90%)                                   | Low        | Gemini Pro      | –                 | Composite 3-term load sum + condition.                              |
| 1D Gibbs Wave PDE Solver                                              | Very High  | Claude Sonnet 5 | Extended Thinking | Implicit FD discretization, BCs, CFL stability. Core module.        |
| Effective Stroke $S_{eff}$ & Displacement                             | Low        | Gemini Flash    | –                 | Direct kinematic algebra chain.                                     |
| $SPM_{safe}(t)$ Speed Floor                                           | Low        | Gemini Flash    | –                 | Inverse viscosity relation formula.                                 |
| Dyno Card Rasterization ( $224 \times 224$ )                          | Med        | Gemini Pro      | –                 | NumPy array to CV tensor pipeline.                                  |
| Diagnostic Classifier (5-State Rule Base)                             | Med        | Gemini Pro      | –                 | Branching condition tree mapping signatures.                        |
| <i>Phase 3 — Wellbore &amp; Surface Transport</i>                     |            |                 |                   |                                                                     |
| Multiphase Flow (Beggs & Brill)                                       | High       | Claude Sonnet 5 | High Effort       | Flow-regime map iterations & holdup equations.                      |
| Thermal-Hydraulic Coupling                                            | Low        | Gemini Pro      | –                 | Wiring/interface code between modules.                              |
| Dynamic Density $\rho_{fluid}(t)$ Feedback                            | Med        | Gemini Pro      | –                 | Dynamic holdup-weighted fluid density.                              |
| <i>Phase 4 — Integrated Well2Surface Optimizer</i>                    |            |                 |                   |                                                                     |
| Economic Objective $J$ + Cut-off Trigger                              | Med–High   | Claude Sonnet 5 | Medium Effort     | Real-time derivative trigger condition ( $dJ/dt \leq 0$ ).          |
| Joint Non-Convex Optimizer (SLSQP/DE)                                 | High       | Claude Sonnet 5 | High Effort       | Constrained 4-var optimization across coupled limits.               |
| <i>Phase 5, 6, 7 &amp; 8 — UI, Exporters, Testing &amp; Core Wire</i> |            |                 |                   |                                                                     |
| Streamlit Dashboard (Dark Theme)                                      | Med        | Gemini Pro      | –                 | Multi-panel UI rendering & plotting layout.                         |
| LaTeX Report Exporter (pylatex) document generation.                  | Low–Med    | Gemini Flash    | –                 | Templated table                                                     |
| CSV / JSON Persistence                                                | Trivial    | Gemini Flash    | –                 | State dump and reloading logic.                                     |
| Unit Tests: Thermal Asymptote                                         | Low        | Gemini Flash    | –                 | Asymptotic limit checks ( $t \rightarrow \infty$ ).                 |
| Unit Tests: CFL Stability & Bounds                                    | Med–High   | Claude Sonnet 5 | Medium Effort     | Validates $\Delta t/\Delta x$ bounds vs wave speed $a$ .            |
| Unit Tests: Edge Cases ( $\mu \rightarrow \infty$ , $SPM = 0$ )       | Low        | Gemini Pro      | –                 | Boundary condition guard tests.                                     |
| <code>main.py</code> Architecture Wiring feedback loop protection.    | Med–High   | Claude Sonnet 5 | Medium Effort     | Data pipeline ordering                                              |
| Latency Benchmark Harness                                             | Low        | Gemini Pro2     | –                 | Profiling & execution timing tests.                                 |

## Phase-by-Phase Technical Requirements

### Phase 0 — Scaffold & Configuration

[Gemini Flash 3.6]

#### Repository Architecture

Build directory tree (config/, data/, src/physics/, src/optimization/, src/dashboard/, src/exporters/, tests/).

#### config/field\_params.yaml

Define Baghewala properties: 18°API,  $T_{\text{res}} = 46\text{--}48^\circ\text{C}$ , petrophysical constants ( $\phi$ ,  $S_o$ ,  $S_w$ ,  $\rho$ ,  $C_p$ ), and Andrade constants ( $A = -14.28$ ,  $B = 5820.4$ ).

#### config/operational\_limits.yaml

Define thresholds: Max rod stress, PPRL yield threshold, min pump fillage %, SPM<sub>safe</sub> floor (2.0 SPM), steam  $T/P$  caps.

### Phase 1 — CSS Thermal EOR Engine

[Claude Sonnet 5] / [Gemini Pro 3.1]

#### Steam Heat Injection (Gemini Pro)

$$Q_{\text{inj}} = m_s [x_{\text{qual}} h_{\text{fg}} + C_w (T_{\text{steam}} - T_{\text{res}})], \quad x_{\text{qual}} \in [0, 1] \quad (1)$$

#### Reservoir Absorption (Gemini Pro)

$$Q_{\text{absorbed}} = V_{\text{zone}} (\rho C)_{\text{composite}} \Delta T, \quad (\rho C)_{\text{composite}} = (1 - \phi) \rho_r C_{pr} + \phi (S_o \rho_o C_{po} + S_w \rho_w C_{pw}) \quad (2)$$

#### Thermal Decay Dynamics (Claude Sonnet 5 — High Effort)

$$\frac{dT}{dt} = -\alpha (T - T_{\text{res}}) \implies T(t) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}}) e^{-\alpha t}, \quad \alpha = \frac{q_o \rho_o C_{po} + q_w \rho_w C_{pw}}{V_{\text{heated}} (\rho C)_{\text{composite}}} \quad (3)$$

#### Viscosity Calibration (Gemini Pro)

Evaluate  $\mu(t) = \exp \left( A + \frac{B}{T(t) + 273.15} \right)$ . Verify  $47^\circ\text{C} \rightarrow \sim 66 \text{ cP}$  and  $250^\circ\text{C} \rightarrow \sim 1 \text{ cP}$ .

### Phase 2 — SRP Mechanical Lift Engine

[Claude Sonnet 5] / [Gemini Pro]

#### Statics & Drag Forces (Gemini Pro / Flash)

Calculate  $\Delta L = \frac{F_{\text{load}} L}{AE}$ ,  $W_{\text{buoy}}$ , and viscous drag:

$$F_{\text{drag}}(t) = \frac{2\pi\mu(t)v_{\text{rod}}L_{\text{rod}}}{\ln(D_{\text{tubing}}/D_{\text{rod}})}, \quad \text{PPRL} = W_r + W_f + F_{\text{drag}} \quad (4)$$

#### 1D Gibbs Wave Equation (Claude Sonnet 5 — Extended Thinking)

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} - c(t) \frac{\partial u}{\partial t}, \quad a = \sqrt{\frac{E}{\rho_{\text{steel}}}} \approx 5000 \text{ m/s}, \quad c(t) = \frac{2\pi\mu(t)}{\rho_{\text{steel}} A_{\text{rod}} \ln(D_{\text{tub}}/D_{\text{rod}})} \quad (5)$$

Discretize via implicit finite-difference scheme with boundary conditions  $u(0, t) = x_{\text{surface}}(t)$  and  $EA \frac{\partial u}{\partial x} \big|_0 = F_{\text{surface}}(t)$  to reconstruct downhole position  $u(L, t)$  and force  $F_{\text{plunger}}(t)$ .

#### Kinematics & Diagnostic Classifier (Gemini Pro)

Evaluate  $\text{SPM}_{\text{safe}}(t) \leq \frac{K(\rho_{\text{steel}} - \rho_{\text{fluid}})g}{\mu(t)}$ , rasterize cards to  $224 \times 224$  tensors, and implement 5-state rule-based diagnostic classifier.

### Phase 3 — Wellbore & Surface Transport

[Claude Sonnet 5] / [Gemini Pro]

#### Multiphase Hydraulics (Claude Sonnet 5 — High Effort)

Implement Beggs & Brill flow correlation with inclination corrections and flow-regime map transitions.

#### Coupling & Density Feedback (Gemini Pro)

Feed dynamic wellbore temperature  $T(t)$  and holdup-weighted fluid density  $\rho_{\text{fluid}}(t)$  back into  $F_{\text{drag}}$ .

## Phase 4 — Integrated Well2Surface Optimizer

[Claude Sonnet 5]

### Economic Objective & Cutoff (Claude Sonnet 5 — Medium Effort)

$$J = \int [P_{\text{oil}} q_o(t) - C_{\text{elec}} P_{\text{motor}}(t) - C_{\text{wear}} \sigma_{\text{rod}}(t)] dt - C_{\text{steam}} m_s \quad (6)$$

Trigger cycle transition when  $\frac{dJ}{dt} \leq 0$  or Steam-Oil Ratio (SOR) exceeds economic threshold.

### Joint Optimization (Claude Sonnet 5 — High Effort)

Multi-variable constrained optimization over  $(m_s, t_{\text{soak}}, \text{SPM}, S)$  using SLSQP/Differential Evolution under dynamic  $\text{SPM}_{\text{safe}}(t)$  and PPRL limits.

## Phase 5 & 6 — Dashboard & Exporters

[Gemini Pro] / [Gemini Flash]

### Streamlit Interface (Gemini Pro)

Dark-theme UI (#121820) featuring wellbore schematic, dual-axis thermal plots, dyno card studio, and VFD auto-throttle console.

### Exporters & Persistence (Gemini Flash)

Auto-generate summary reports via `pylatex`, CSV time-series logs, and JSON state serialization.

## Phase 7 & 8 — Verification & Integration

[Claude Sonnet 5] / [Gemini Pro]

### CFL Stability Unit Test (Claude Sonnet 5 — Medium Effort)

Verify finite-difference mesh stability condition ( $\Delta t \leq \Delta x/a$ ).

### System Architecture Integration (Claude Sonnet 5 — Medium Effort)

Wire end-to-end data pipeline across all 5 modules with fault-tolerant boundary validation.

### Edge Case Testing & Latency Benchmark (Gemini Pro)

Test cold shutdown ( $\mu \rightarrow \infty$ ), zero stroke, and verify wave solver execution latency ( $< 200$  ms).